

## COE817 Lab 3

### Objectives

- In this lab, you will work on a key distribution protocol. Netbeans is the IDE.
- **Two students** will work as a group.
- Duration: **one week**.

### Your tasks

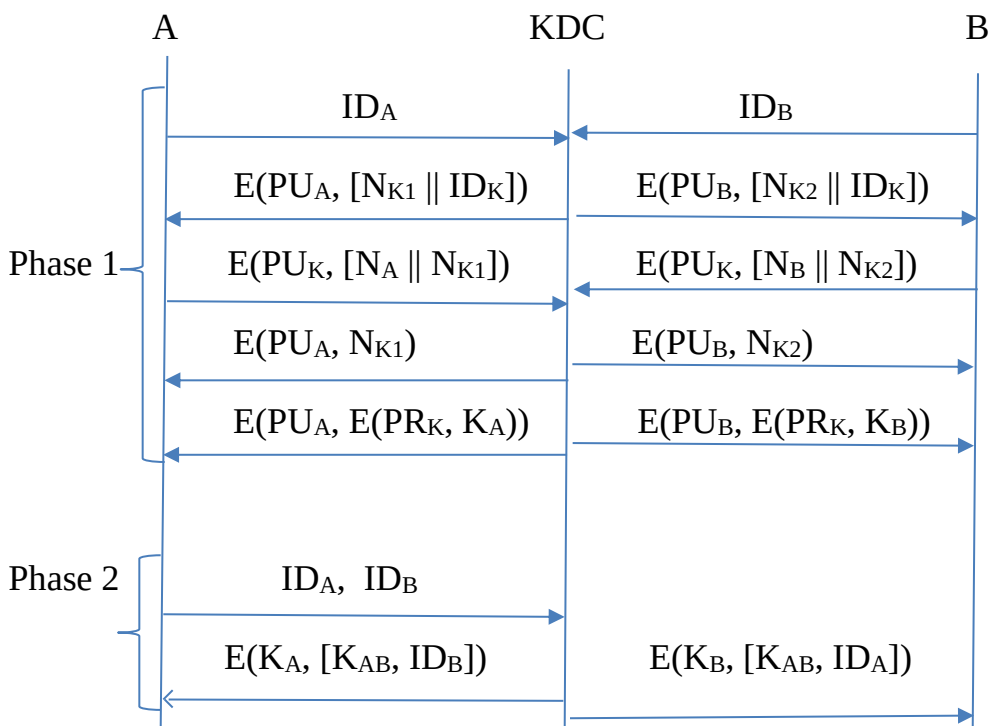
Write a Java socket program to implement a hybrid key distribution protocol that is explained in class. A server (KDC) is needed for exchanging messages with two clients (A and B).

The key distribution has the following two phases:

Phase 1: Public-key scheme is used to distribute a master key  $K_A$  (a symmetric key shared key by KDC and client A) and a master key  $K_B$  (a symmetric key shared by KDC and client B). Both  $K_A$  and  $K_B$  are generated by KDC.

Phase 2: The server (KDC) then generated a session key  $K_{AB}$  and separately send it to A and B

Implement the key distribution protocol as diagrammed below.



PU<sub>A</sub>: Public key of A  
PU<sub>B</sub>: Public key of B  
PU<sub>K</sub>: Public key of KDC  
PR<sub>K</sub>: Private key of KDC  
N<sub>K1</sub>, N<sub>K2</sub>: nonces generated by the KDC  
ID<sub>K</sub>: ID of the KDC  
K<sub>A</sub>: master key of A and KDC  
K<sub>B</sub>: master key of B and KDC  
K<sub>AB</sub>: Session key of A, B

You need to

- (1) Generate RSA public/private key pairs for KDC, A and B
- (2) Encrypt and decrypt messages using RSA
- (3) Pick up IDs, K<sub>A</sub>, K<sub>B</sub>, K<sub>AB</sub> and nonces by yourself.

## **What you needed to demonstrate to your TA**

For Phase 1 of the project, show TA:

- (1) The public and private key pairs of A, B and the server
- (2) The decrypted K<sub>A</sub> and K<sub>B</sub>

For Phase 2, demonstrate if both A and B could correctly receive the same K<sub>AB</sub> from the server. Explain your code and tell how your server socket program make it work.  
The protocol is vulnerable. why? How to solve the problem?

## **Submitting your lab**

You must submit your source codes to lab3 submission folder on D2L at least 24 hours prior to the start of your scheduled lab period for Lab4.